

Auteur: Siebe Haché
Studierichting: Informatica
Oefeningenreeks 4A nr. 38.9

$$\begin{aligned}& \int \frac{x^2 + 6x + 12}{x^2 + 4x + 5} dx \\&= \int \frac{(x^2 + 4x + 5) + 2x + 7}{x^2 + 4x + 5} dx \\&= \int \left(1 + \frac{2x + 7}{x^2 + 4x + 5} \right) dx \\&= \int dx + \int \frac{(2x + 4) + 3}{x^2 + 4x + 5} dx \\&= \int dx + \int \frac{2x + 4}{x^2 + 4x + 5} dx + 3 \int \frac{dx}{x^2 + 4x + 5} \\&u = x^2 + 4x + 5 \Rightarrow du = (2x + 4)dx \\&v = x + 2 \Rightarrow dv = dx \\&= \int dx + \int \frac{du}{u} + 3 \int \frac{dv}{v^2 + 1} \\&= x + \ln |u| + 3 \operatorname{Bgtan} v + C \\&= x + \ln |x^2 + 4x + 5| + 3 \operatorname{Bgtan}(x + 2) + C\end{aligned}$$

References